Lung Cancer Detection Using Convolutional Neural Networks

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Course: PBEL 2.0 AI

Batch: 8th

College: Mody University of Science and Technology

Submitted To: Mr. Devansh

Introduction & Problem Statement

The Challenge

Lung cancer is the leading cause of cancer mortality worldwide, with over 2 million new cases annually. Early detection dramatically improves survival rates, but traditional diagnosis requires expert radiologists and is time-consuming.

AI Solution

This project addresses these challenges by developing an AI-based system that automatically analyzes lung CT scan images using Convolutional Neural Networks (CNNs), assisting radiologists with faster, more accurate diagnosis.

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Project Objectives

Deep Learning Model

Build a CNN architecture to classify lung CT scan images as cancerous or normal with high accuracy

Cancer Detection

Develop an automated system to detect presence of lung cancer from medical imaging data

Web Deployment

Deploy the trained model as an accessible web-based application for practical medical use

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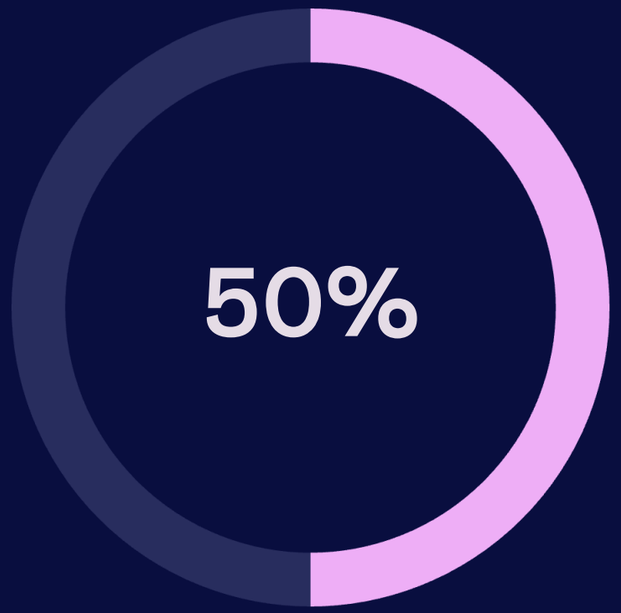
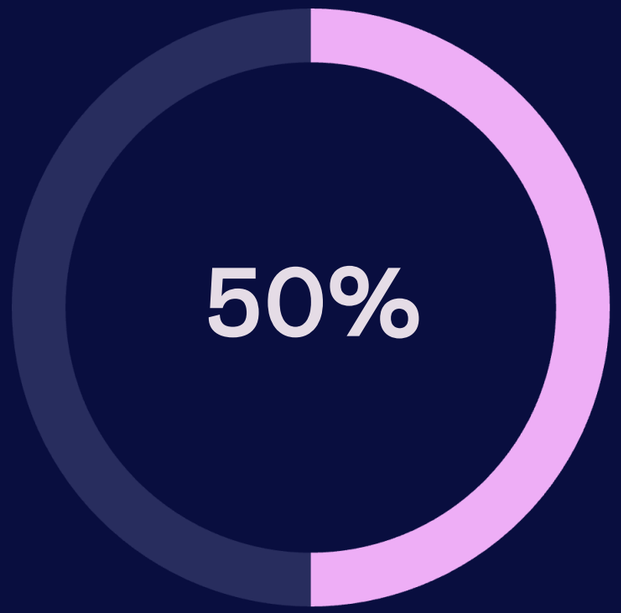
Dataset Overview

The **IQ-OTHNCCD lung cancer dataset** contains CT scan images categorized into three classes:



Cancer Class

Benign and malignant cases merged into single cancer category for binary classification



Normal Class

Healthy lung CT scans with no cancerous tissue present

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Data Preprocessing Pipeline

Image Resizing

Standardize all images to 128×128 pixels for consistent model input

Normalization

Scale pixel values to [0,1] range for improved neural network training

Dataset Splitting

Divide data into training (80%) and testing (20%) sets

Data Augmentation

Generate additional training samples through rotations, flips, and transformations

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CNN Architecture

01

Convolutional Layers

Extract features through multiple filter applications

02

MaxPooling

Reduce spatial dimensions while retaining important features

03

Flatten Layer

Convert 2D feature maps into 1D vector for classification

04

Dense Layers

Perform classification using learned features

05

Dropout

Prevent overfitting by randomly disabling neurons during training

06

Sigmoid Output

Generate probability prediction for binary classification

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Model Training & Evaluation

Training Configuration

- **Loss Function:** Binary cross-entropy for binary classification
- **Optimizer:** Adam optimizer for adaptive learning rates
- **Epochs:** 50 training iterations with batch size of 32
- **Validation:** Monitor performance on test set during training

Evaluation Metrics

- **Accuracy:** Overall correctness of predictions
- **Confusion Matrix:** Detailed breakdown of true/false positives and negatives
- **Validation Performance:** Generalization to unseen data

System Implementation

Image Upload

CT scan image uploaded through Streamlit web interface

Preprocessing

Image resized, normalized, and formatted for model input

CNN Analysis

Trained model processes image and extracts features

Prediction Output

System displays diagnosis with confidence score

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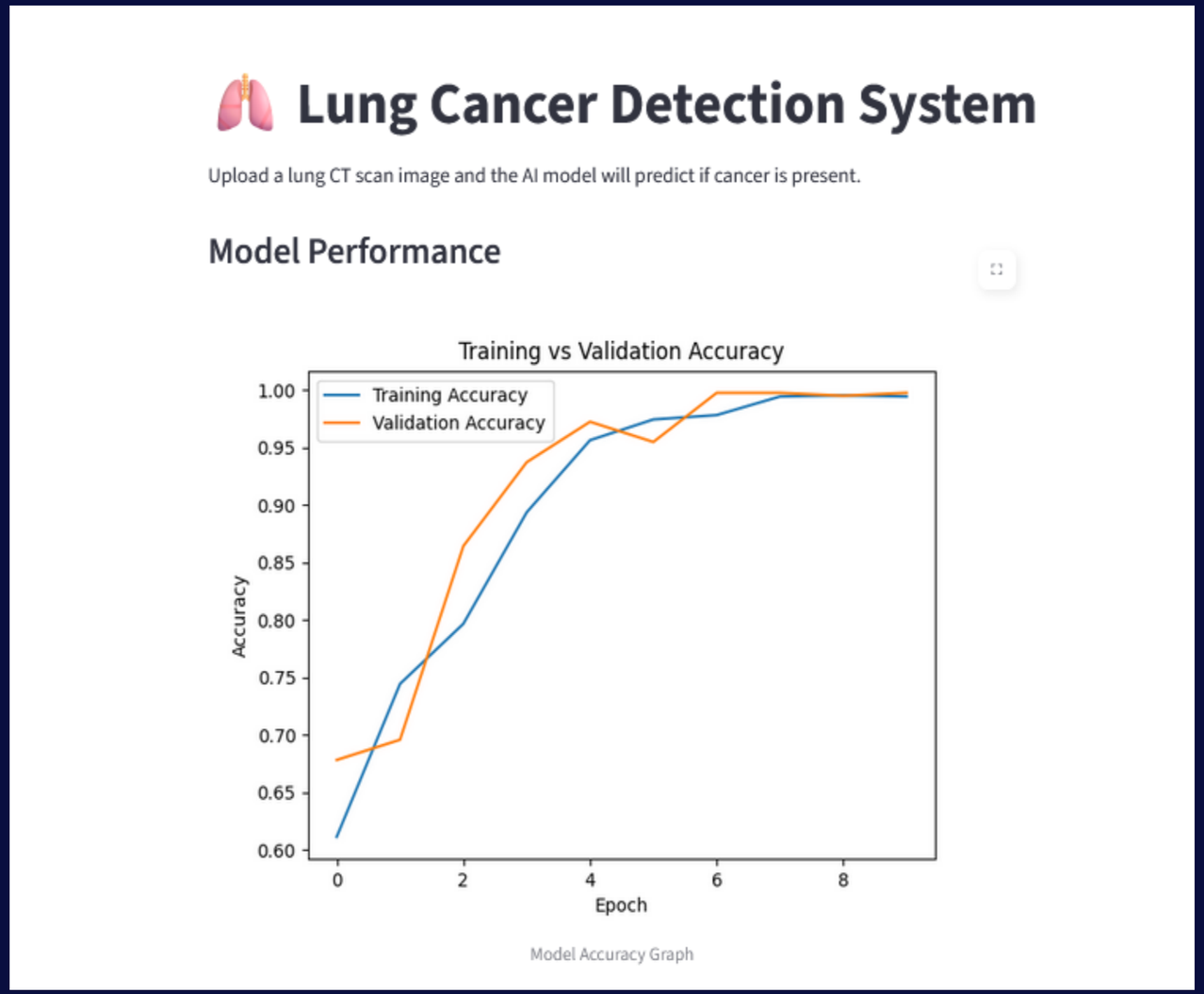
Result:-



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Challenges & Future Improvements

Current Challenges

- Limited dataset size affecting model robustness
- Slice-based CT analysis rather than 3D volume
- Potential overfitting to training data

Future Enhancements

- Integration with larger medical imaging datasets
- Transfer learning using pre-trained models (ResNet, VGG16)
- 3D CNN architectures for volumetric analysis
- Clinical integration and validation studies

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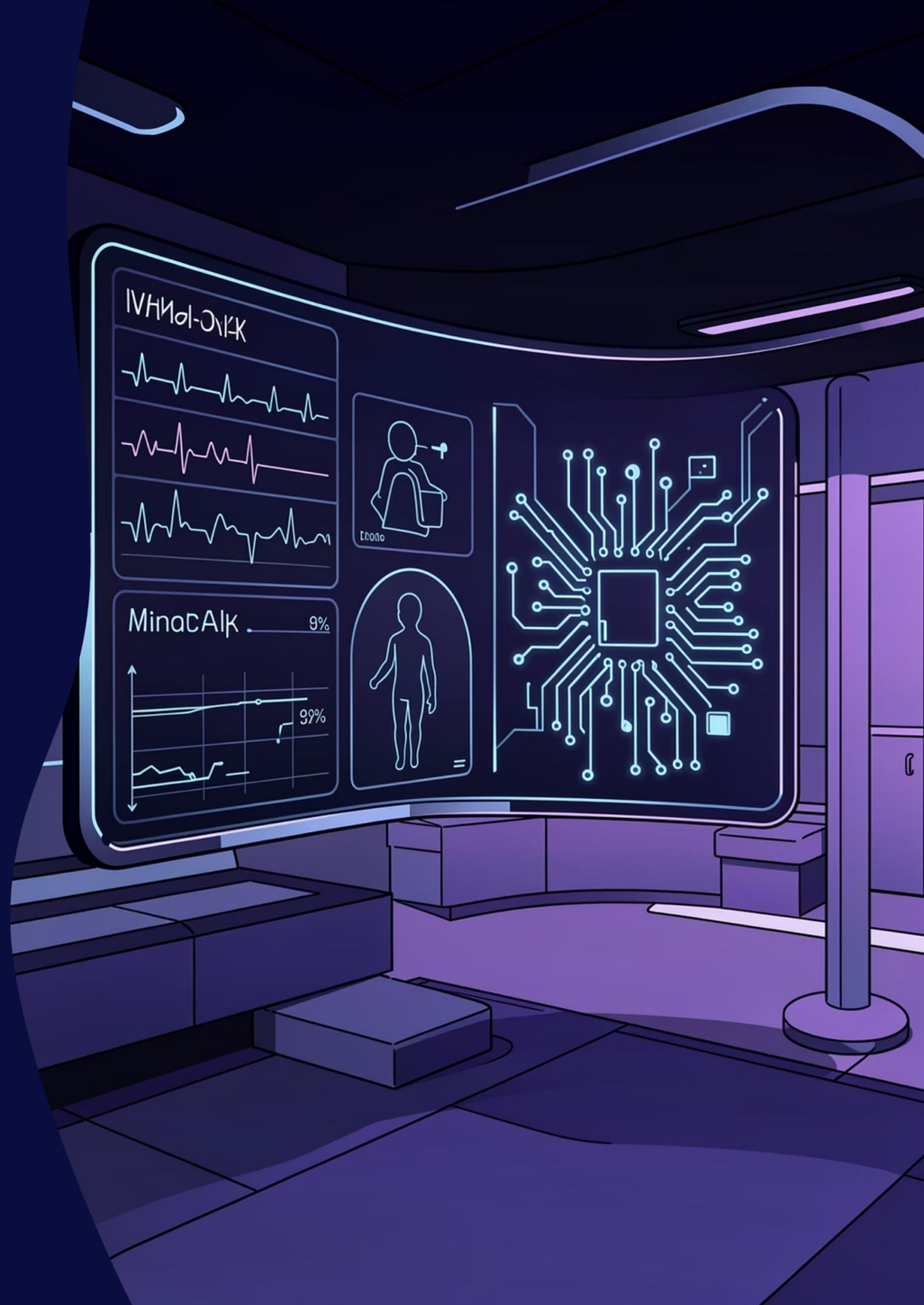
Thankyou